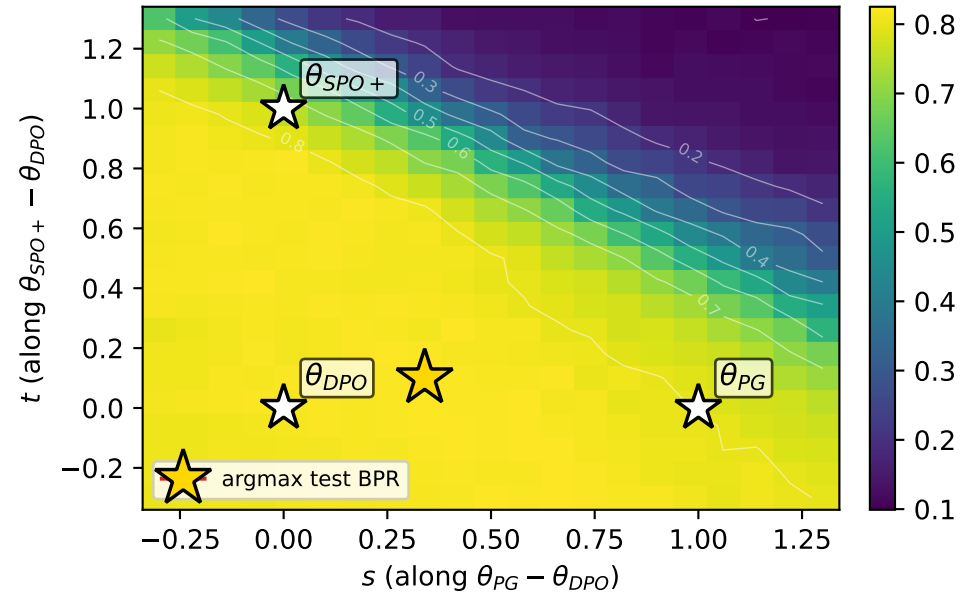


Cook County IL Fatal Overdoses (K=100)

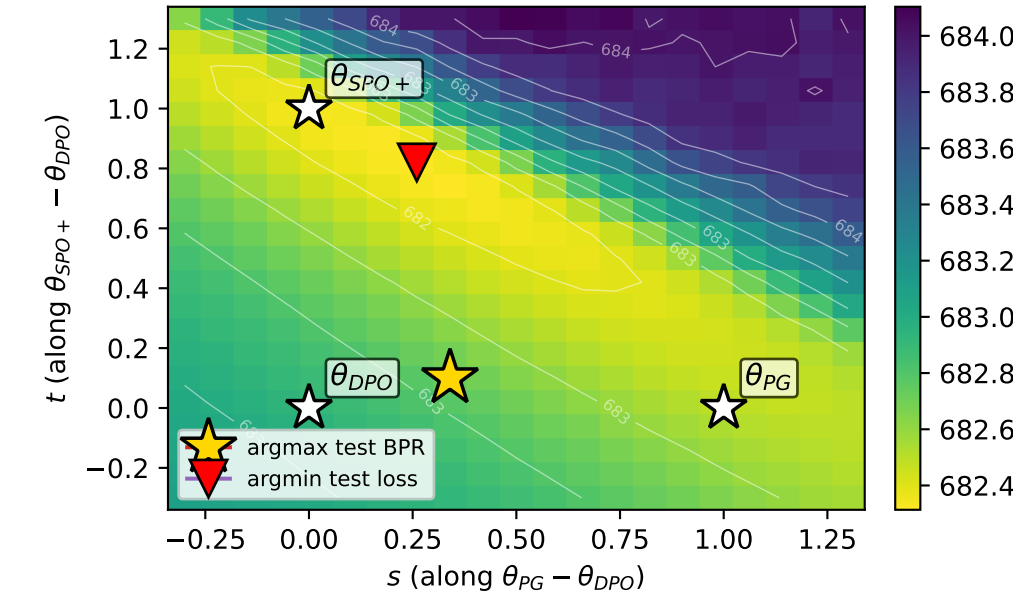
$$\theta(s, t) = (1 - s - t) \theta_{DPO} + s \theta_{PG} + t \theta_{SPO+}$$

Test BPR (higher = better)



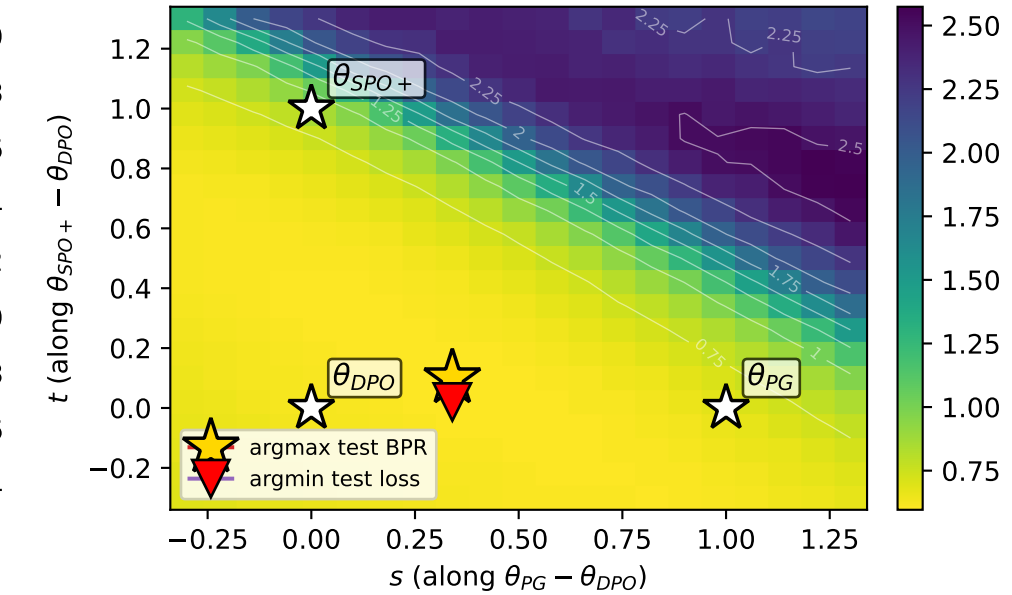
SPO+ surrogate on decision loss

true cost = y_t ; upper-bounds regret on $\max y_t^\top z_t$ (lower = better)
corr(test BPR, -loss) = +0.903



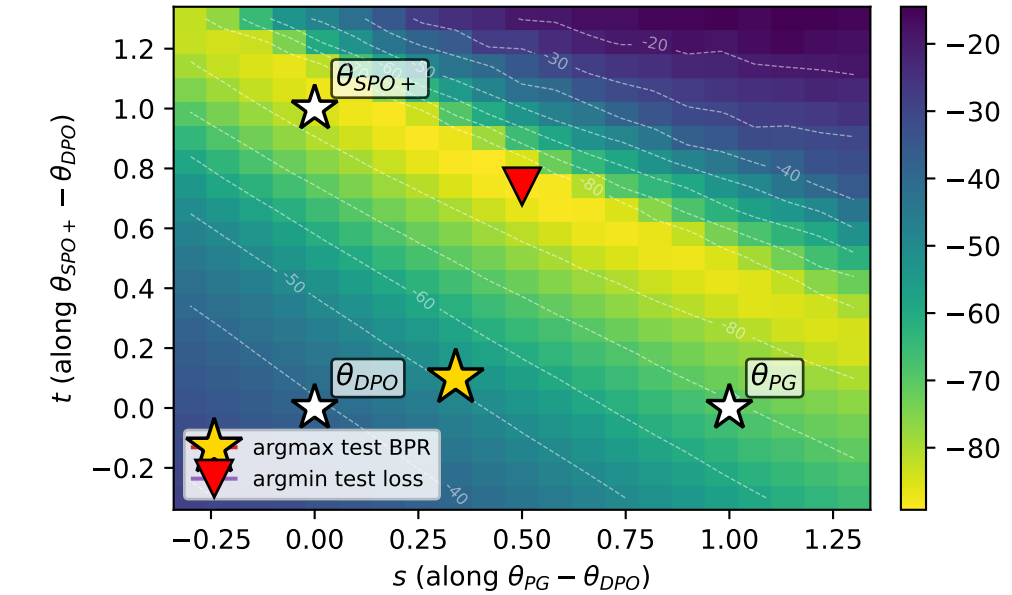
SPO+ surrogate on BPR

true cost = y_t/C_t ; upper-bounds regret on $\max y_t^\top z_t/C_t$ (lower = better)
corr(test BPR, -loss) = +0.985



PG surrogate on decision loss @ $h=0.01$

true cost = y_t ; $\text{PGB} \rightarrow \mathbb{E}_t[-y_t^\top z_t]$ as $h \rightarrow 0$, $z_t = \text{top-}K$ of t_t (lower = better)
corr(test BPR, -loss) = +0.483



PG surrogate on BPR @ $h=0.01$

true cost = y_t/C_t ; $\text{PGB} \rightarrow \mathbb{E}_t[-y_t^\top z_t/C_t] = \mathbb{E}_t[-\text{BPR}_t]$ as $h \rightarrow 0$
corr(test BPR, -loss) = +1.000

